

Title: GARUDA AI — An AI-Driven Crime Pattern Analysis and Public Safety Intelligence System

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Domain

Public Safety Analytics / Crime Intelligence (Data Science & Big Data Analytics)

Dataset

I recommend using three benchmark datasets together:

1. UCI Communities and Crime Dataset (socio-economic, demographic, and law-enforcement factors linked to crime rates)

<https://archive.ics.uci.edu/dataset/183/communities+and+crime>

2. Chicago Crime Dataset — 2001 to Present (large-scale, real-world reported incident data with location, time, and crime type)

<https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2>

3. Crime in India Dataset (NCRB) (state/district-wise Indian crime records across 40+ crime categories, for regional/local relevance)

<https://www.kaggle.com/datasets/rajanand/crime-in-india>

Abstract

Urban law enforcement agencies generate large volumes of crime, demographic, and geospatial data, yet conventional systems remain largely reactive and predictive models often operate as black boxes. GARUDA AI addresses this gap through an Explainable AI (XAI)-based framework for Crime Pattern Analysis and Public Safety Intelligence. It integrates crime data mining, spatial analytics, EDA, multi-model machine learning, and explainability using three benchmark datasets: UCI Communities and Crime, Chicago Crime (2001–Present), and NCRB Crime in India. The datasets undergo missing-value handling, encoding, geocoding, normalization, duplicate and outlier detection, descriptive, univariate, bivariate and multivariate analysis, correlation analysis, feature engineering, and feature selection.

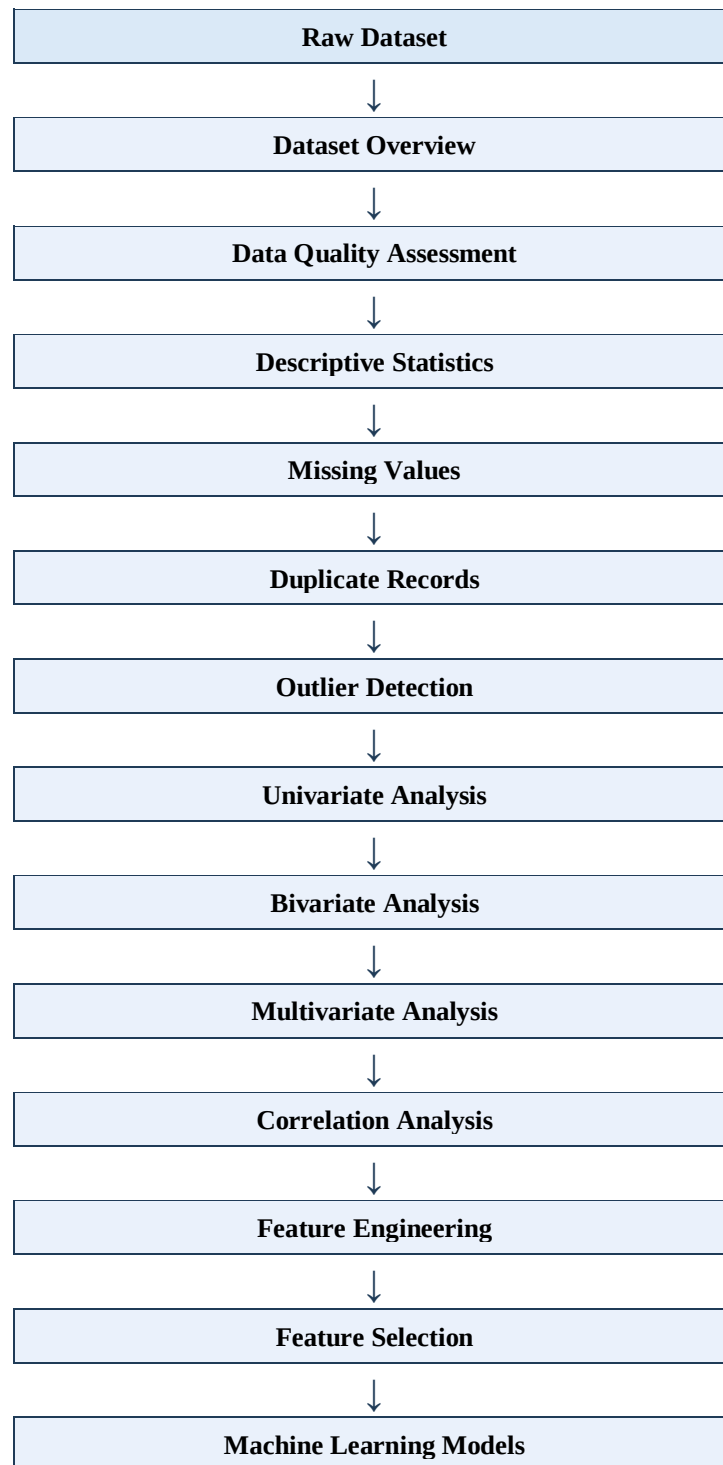
For prediction, Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost are used as core models. Their predictions are independently cross-verified through Isolation Forest for anomaly detection, LSTM for temporal consistency, and DBSCAN and K-Means for spatial hotspot verification. A prediction is considered strongly verified when these classifier, statistical, temporal, and spatial perspectives agree. Model reliability is evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, k-fold cross-validation, and cross-model agreement, ensuring performance is not judged by a single metric or train-test split.

To ensure transparency, GARUDA AI combines SHAP and LIME for global and local explanations, Feature Importance for key predictors, Partial Dependence for feature–risk relationships, and counterfactual/anchor explanations to show why predictions change. Results are presented through heatmaps, boxplots, feature-importance graphs, hotspot maps, and an interactive dashboard. Comparative analysis identifies the most accurate, cross-verified, and explainable approach.

Approach:

Data Analytics

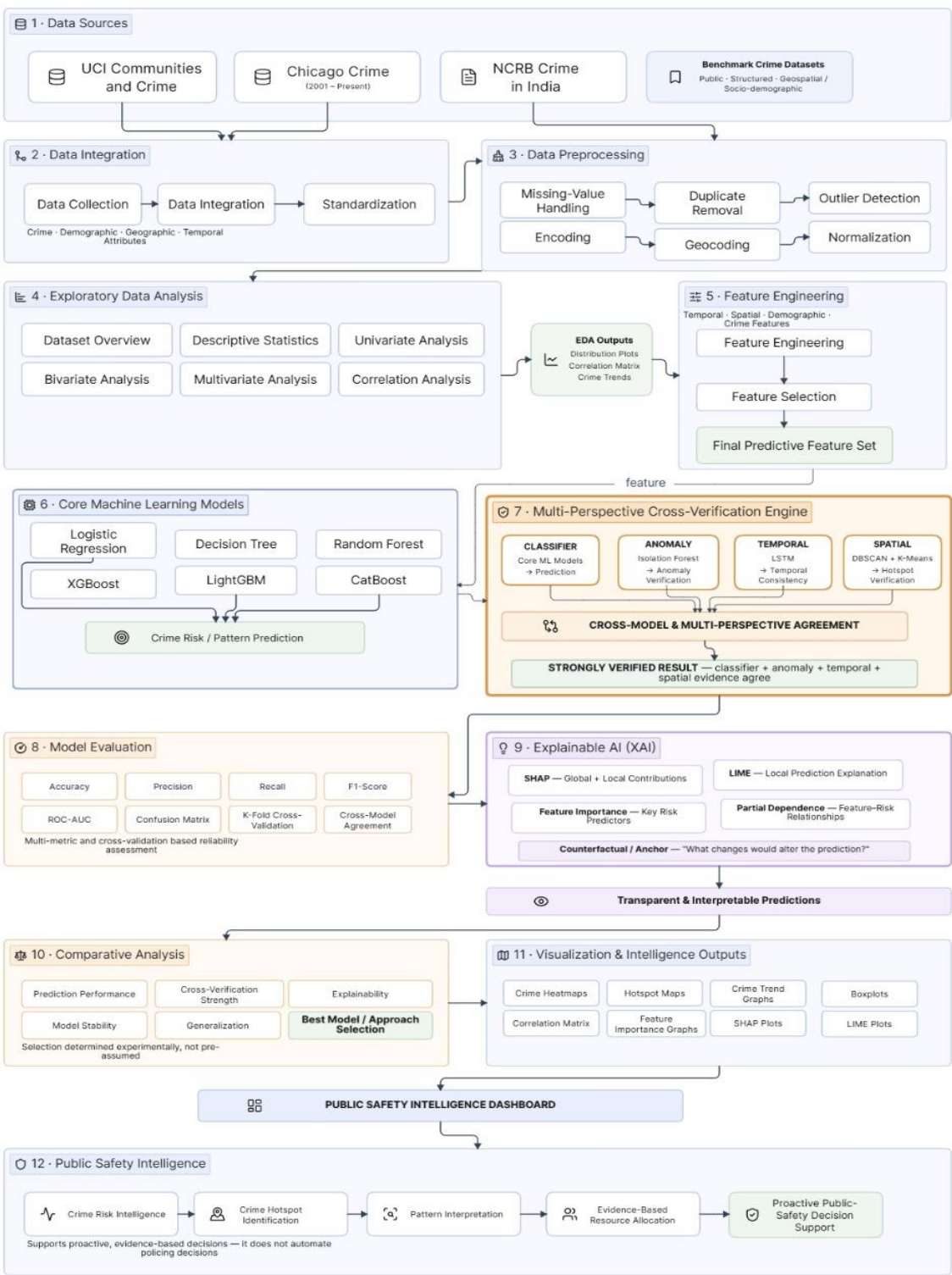
- EDA
- EDA Flow Diagram:



ARCHITECTURE DIAGRAM :-

GARUDA AI AI-Driven Crime Pattern Analysis and Public Safety Intelligence System

Explainable AI Framework for Crime Pattern Analysis and Public Safety Intelligence



Preprocessing

- Missing values
- Categorical encoding
- Normalization
- Geocoding (mapping incident locations to latitude/longitude for spatial analysis)

ML Models

Core predictive classifiers:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost

Independent cross-verification layers:

- Isolation Forest — anomaly-based verification (is this incident a genuine statistical outlier?)
- LSTM — temporal verification (is this consistent with the area's historical trend over time?)
- DBSCAN / K-Means — spatial verification (does this fall within a real geographic hotspot, not an isolated point?)

Explainability

- SHAP — global and local feature attribution
- LIME — local, instance-level explanation
- Feature Importance ranking
- Partial Dependence analysis
- Counterfactual / Anchor-based explanations

Visualization

- Heatmap
- Boxplot
- Feature Importance
- Hotspot / GIS Map
- Dashboard

Comparative Analysis

Compare

- Decision Tree
- Random Forest
- XGBoost
- LightGBM
- CatBoost
- Isolation Forest (anomaly-verification layer)
- LSTM (temporal-verification layer)

Metrics

- Accuracy & Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix
- k-Fold Cross-Validation

Recommended Research Pipeline:

